

OpenStax Calculus Volume 2

Section 7.4 Exercises: Area and Arc Length in Polar Coordinates

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For the following exercises, determine a definite integral that represents the area.

1. Region enclosed by $r = 4$
2. Region enclosed by $r = 3 \sin \theta$
3. Region in the first quadrant within the cardioid $r = 1 + \sin \theta$
4. Region enclosed by one petal of $r = 8 \sin(2\theta)$
5. Region enclosed by one petal of $r = \cos(3\theta)$
6. Region below the polar axis and enclosed by $r = 1 - \sin \theta$
7. Region in the first quadrant enclosed by $r = 2 - \cos \theta$
8. Region enclosed by the inner loop of $r = 2 - 3 \sin \theta$
9. Region enclosed by the inner loop of $r = 3 - 4 \cos \theta$
10. Region enclosed by $r = 1 - 2 \cos \theta$ and outside the inner loop
11. Region common to $r = 3 \sin \theta$ and $r = 2 - \sin \theta$
12. Region common to $r = 2$ and $r = 4 \cos \theta$
13. Region common to $r = 3 \cos \theta$ and $r = 3 \sin \theta$

For the following exercises, find the area of the described region.

14. Enclosed by $r = 6 \sin \theta$
15. Above the polar axis enclosed by $r = 2 + \sin \theta$
16. Below the polar axis and enclosed by $r = 2 - \cos \theta$
17. Enclosed by one petal of $r = 4 \cos(3\theta)$
18. Enclosed by one petal of $r = 3 \cos(2\theta)$
19. Enclosed by $r = 1 + \sin \theta$
20. Enclosed by the inner loop of $r = 3 + 6 \cos \theta$

21. Enclosed by $r = 2 + 4 \cos \theta$ and outside the inner loop
22. Common interior of $r = 4 \sin(2\theta)$ and $r = 2$
23. Common interior of $r = 3 - 2 \sin \theta$ and $r = -3 + 2 \sin \theta$
24. Common interior of $r = 6 \sin \theta$ and $r = 3$
25. Inside $r = 1 + \cos \theta$ and outside $r = \cos \theta$
26. Common interior of $r = 2 + 2 \cos \theta$ and $r = 2 \sin \theta$

For the following exercises, find a definite integral that represents the arc length.

27. $r = 4 \cos \theta$ on the interval $0 \leq \theta \leq \frac{\pi}{2}$
28. $r = 1 + \sin \theta$ on the interval $0 \leq \theta \leq 2\pi$
29. $r = 2 \sec \theta$ on the interval $0 \leq \theta \leq \frac{\pi}{3}$
30. $r = e^\theta$ on the interval $0 \leq \theta \leq 1$

For the following exercises, find the length of the curve over the given interval.

31. $r = 6$ on the interval $0 \leq \theta \leq \frac{\pi}{2}$
32. $r = e^{3\theta}$ on the interval $0 \leq \theta \leq 2$
33. $r = 6 \cos \theta$ on the interval $0 \leq \theta \leq \frac{\pi}{2}$
34. $r = 8 + 8 \cos \theta$ on the interval $0 \leq \theta \leq \pi$
35. $r = 1 - \sin \theta$ on the interval $0 \leq \theta \leq 2\pi$

For the following exercises, use the integration capabilities of a calculator to approximate the length of the curve.

36. [T] $r = 3\theta$ on the interval $0 \leq \theta \leq \frac{\pi}{2}$
37. [T] $r = \frac{2}{\theta}$ on the interval $\pi \leq \theta \leq 2\pi$
38. [T] $r = \sin^2(\frac{\theta}{2})$ on the interval $0 \leq \theta \leq \pi$
39. [T] $r = 2\theta^2$ on the interval $0 \leq \theta \leq \pi$
40. [T] $r = \sin(3 \cos \theta)$ on the interval $0 \leq \theta \leq \pi$

For the following exercises, use the familiar formula from geometry to find the area of the region described and then confirm by using the definite integral.

41. $r = 3 \sin \theta$ on the interval $0 \leq \theta \leq \pi$
42. $r = \sin \theta + \cos \theta$ on the interval $0 \leq \theta \leq \pi$
43. $r = 6 \sin \theta + 8 \cos \theta$ on the interval $0 \leq \theta \leq \pi$

For the following exercises, use the familiar formula from geometry to find the length of the curve and then confirm using the definite integral.

44. $r = 3 \sin \theta$ on the interval $0 \leq \theta \leq \pi$
45. $r = \sin \theta + \cos \theta$ on the interval $0 \leq \theta \leq \pi$
46. $r = 6 \sin \theta + 8 \cos \theta$ on the interval $0 \leq \theta \leq \pi$
47. Verify that if $y = r \sin \theta = f(\theta) \sin \theta$, then $\frac{dy}{d\theta} = f'(\theta) \sin \theta + f(\theta) \cos \theta$.

For the following exercises, find the slope of a tangent line to a polar curve $r = f(\theta)$. Let $x = r \cos \theta = f(\theta) \cos \theta$ and $y = r \sin \theta = f(\theta) \sin \theta$, so the polar equation $r = f(\theta)$ is now written in parametric form.

48. Use the definition of the derivative $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$ and the product rule to derive the derivative of a polar equation.
49. $r = 1 - \sin \theta$; $(\frac{1}{2}, \frac{\pi}{6})$
50. $r = 4 \cos \theta$; $(2, \frac{\pi}{3})$
51. $r = 8 \sin \theta$; $(4, \frac{5\pi}{6})$
52. $r = 4 + \sin \theta$; $(3, \frac{3\pi}{2})$
53. $r = 6 + 3 \cos \theta$; $(3, \pi)$
54. $r = 4 \cos(2\theta)$; tips of the leaves
55. $r = 2 \sin(3\theta)$; tips of the leaves
56. $r = 2\theta$; $(\frac{\pi}{2}, \frac{\pi}{4})$
57. Find the points on the interval $-\pi \leq \theta \leq \pi$ at which the cardioid $r = 1 - \cos \theta$ has a vertical or horizontal tangent line.
58. For the cardioid $r = 1 + \sin \theta$, find the slope of the tangent line when $\theta = \frac{\pi}{3}$.

For the following exercises, find the slope of the tangent line to the given polar curve at the point given by the value of θ .

59. $r = 3 \cos \theta$, $\theta = \frac{\pi}{3}$
60. $r = \theta$, $\theta = \frac{\pi}{2}$
61. $r = \ln \theta$, $\theta = e$
62. [T] Use technology: $r = 2 + 4 \cos \theta$ at $\theta = \frac{\pi}{6}$

For the following exercises, find the points at which the following polar curves have a horizontal or vertical tangent line.

63. $r = 4 \cos \theta$
64. $r^2 = 4 \cos(2\theta)$
65. $r = 2 \sin(2\theta)$
66. The cardioid $r = 1 + \sin \theta$
67. Show that the curve $r = \sin \theta \tan \theta$ (called a cissoid of Diocles) has the line $x = 1$ as a vertical asymptote.